Domain Name System (or Service) (DNS)

Computer Networks
Term B10

DNS Outline

- DNS Hierarchial Structure
- Root Name Servers
- Top-Level Domain Servers
- Authoritative Name Servers
- Local Name Server
- Caching and Updating DNS Records
- DNS Protocols and Messages

DNS: Domain Name System

People: many identifiers:

- SSN, name, passport #

Internet hosts, routers:

- IP address (32 bit) - used for addressing datagrams
- "name", e.g., `www.yahoo.com` - used by humans

Q: map between IP addresses and name?

Domain Name System:

- distributed database implemented in hierarchy of many name servers
- application-layer protocol host, routers, name servers to communicate to resolve names (address/name translation)
  - note: core Internet function, implemented as application-layer protocol
  - complexity at network's "edge"

DNS

- DNS servers often run on Unix machines running BIND (Berkeley Internet Name Domain)
- Runs over UDP
- Uses port 53

DNS

DNS services

- hostname to IP address translation
- host aliasing
  - Aliases, where canonical name is "real" name
- mail server aliasing
- load distribution
  - replicated Web servers: set of IP addresses for one name

Why not centralize DNS?

- single point of failure
- traffic volume
- distant centralized database
- maintenance

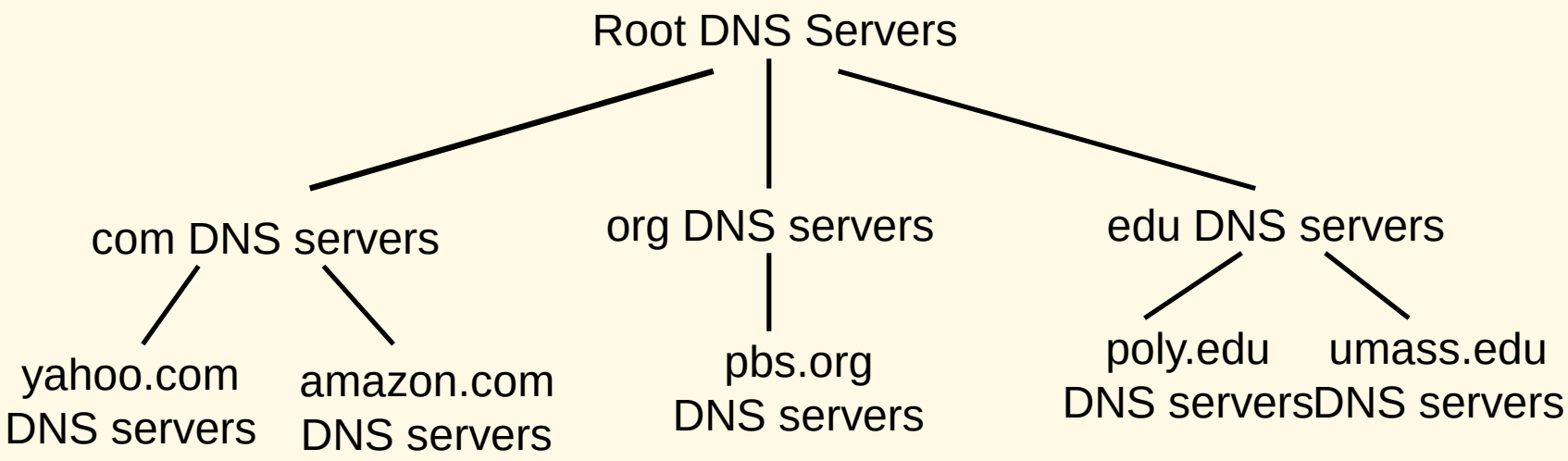
→doesn't *scale!*

- Distributed by design

DNS

- Three classes of servers (approximation):
  - Root DNS servers
  - Top-level domain (TLD) servers
  - Authoritative name servers
- Additionally:
  - Local name server

Distributed, Hierarchical Database

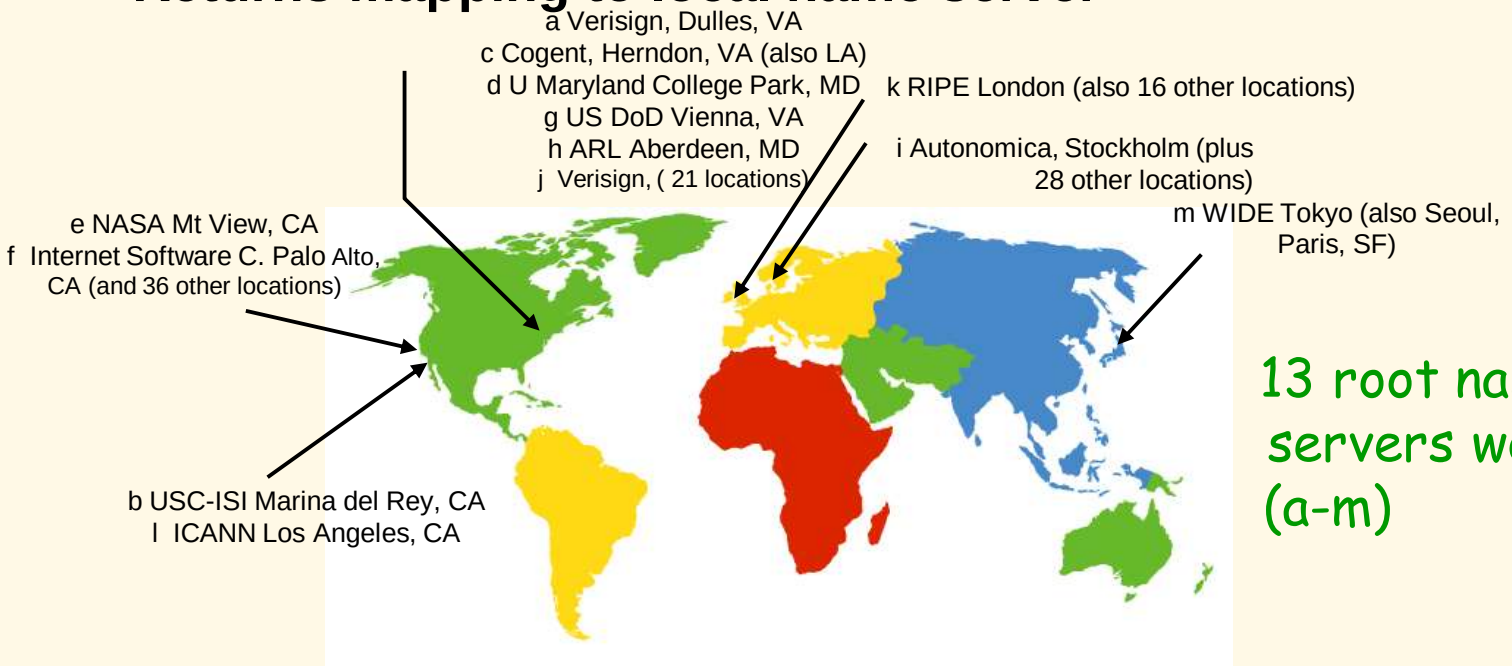


Example: Client wants IP for `www.amazon.com` {1st approx:}

- client queries a root server to find `.com` DNS server
- client queries `.com` DNS server to get `amazon.com` DNS server
- client queries `amazon.com` DNS server to get IP address for `www.amazon.com`

DNS: Root Name Servers

- Contacted by local name server that can not resolve name
- Root name server:
  - Contacts authoritative name server if name mapping not known
  - Gets mapping
  - Returns mapping to local name server



13 root name servers worldwide (a-m)

Top-Level Domain (TLD)

- Top-level domain (TLD) servers:
  - Responsible for com, org, net, edu, etc, and all top-level country domains such as uk, fr, ca and jp.
  - Network Solutions maintains servers for com TLD.
  - Educause for edu TLD.
  - VeriSign for net TLD.

Authoritative Servers

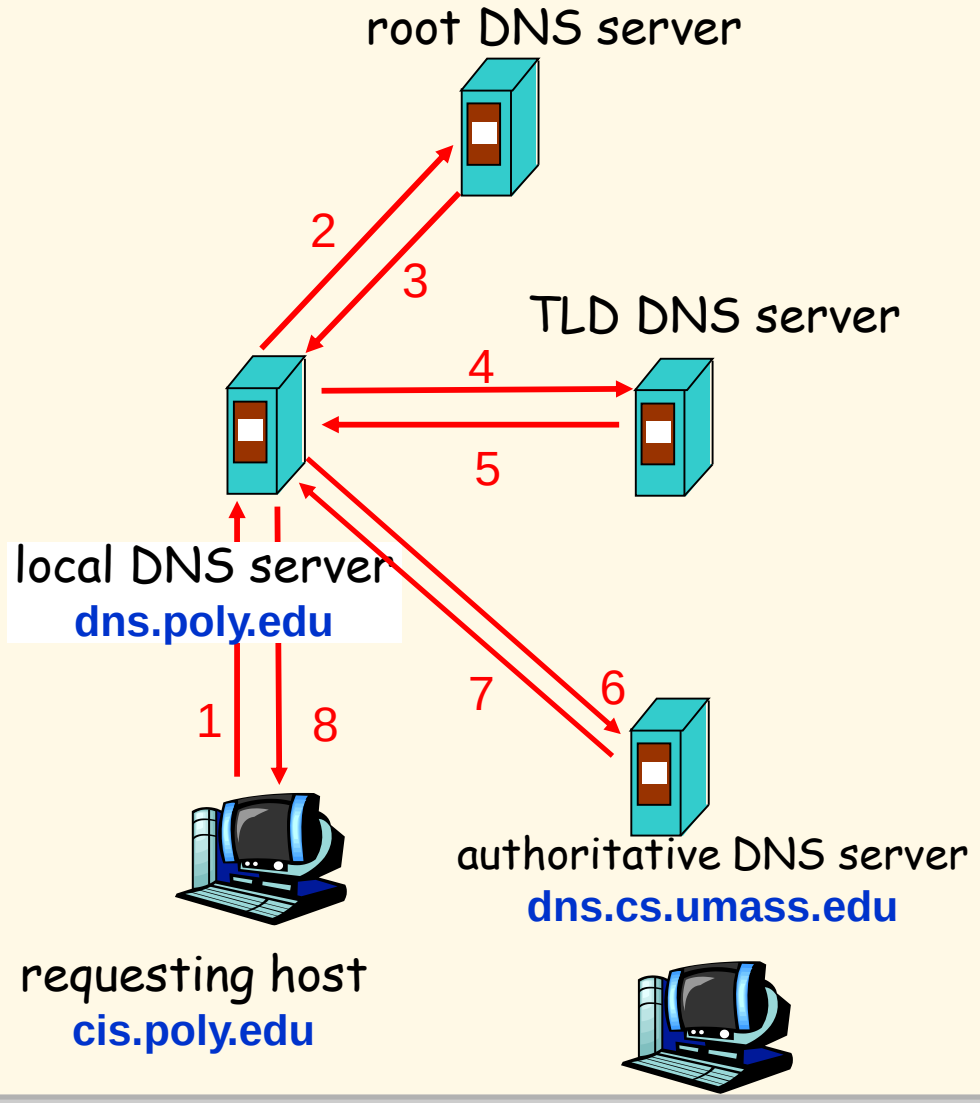
- Authoritative DNS servers:
  - Organization's DNS servers, providing authoritative hostname to IP mappings for organization's servers (e.g., Web, mail).
  - Can be maintained by organization or service provider.

Local Name Server

- Does not strictly belong to hierarchy.
- Each ISP (residential ISP, company, university) has one
  - Also called "default name server"
  - You can run one in your home/dorm!
- When a host makes a DNS query, the query is sent to its local DNS server.
  - ISP provides IP address of local DNS server using DHCP.
  - Acts as proxy, forwards query into the name server hierarchy.

DNS Name Resolution Example

- **Host at `cis.poly.edu` wants IP address for `gaia.cs.umass.edu`**



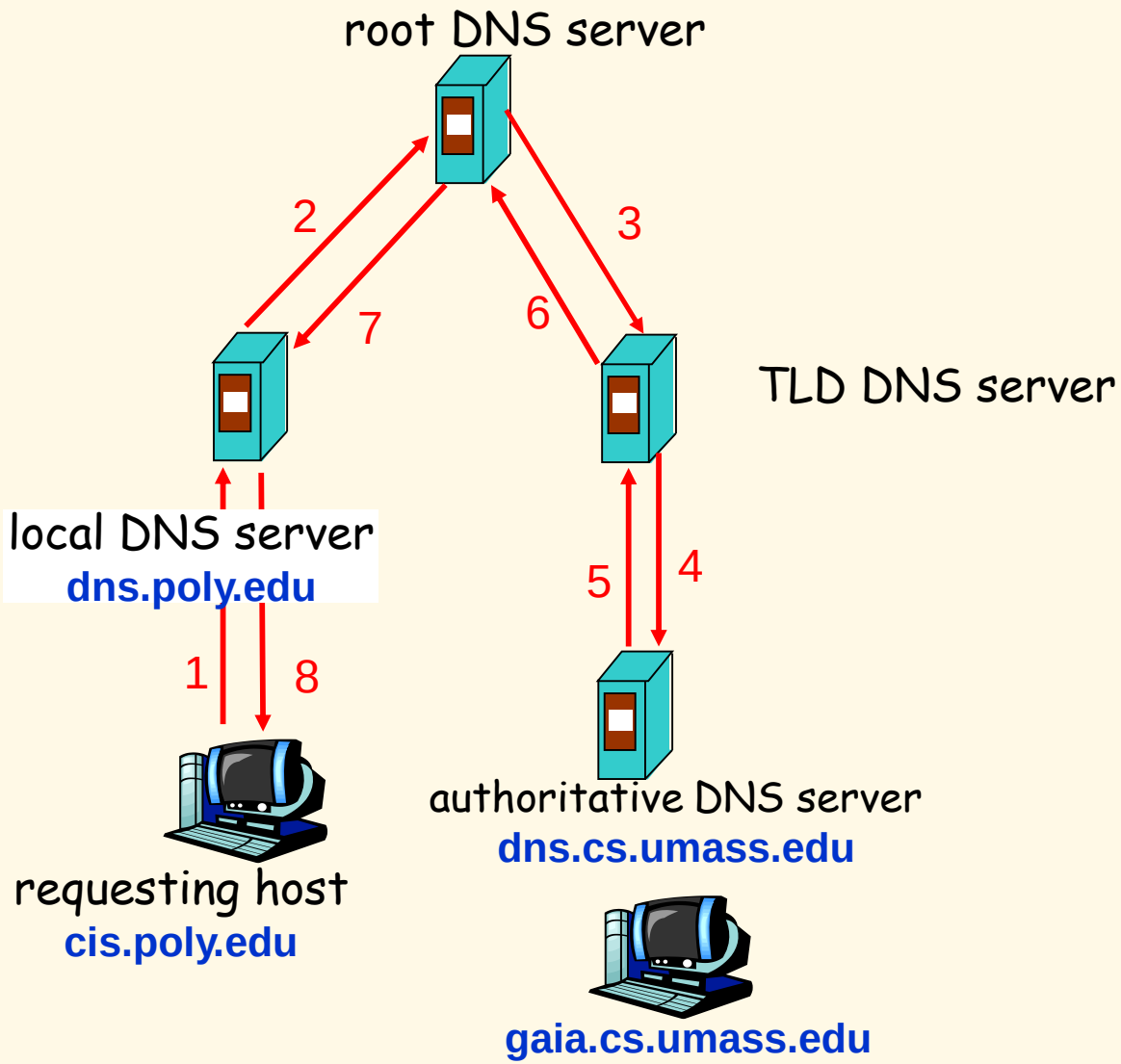
Iterated query

- contacted server replies with name of server to contact.
- "I don't know this name, but ask this server."

DNS Name Resolution (example)

Recursive query

- Puts burden of name resolution on contacted name server.
- Heavy load?



DNS: Caching and Updating Records

- Once (any) name server learns mapping, it *caches* mapping.
  - Cache entries timeout (disappear) after some time (e.g two days) {specified as TTL ==Time-To-Live}.
  - IP addresses of TLD servers are typically cached in local name servers.
    - Thus root name servers are not visited frequently.
- Originally thought DNS names quite static, but increasingly not so → update/notify mechanisms under design by IETF
  - RFC 2136: http://www.ietf.org/rfc/rfc2136.txt

DNS Records

DNS: distributed database storing resource records (RR)

RR format: `(name, value, type, ttl)`

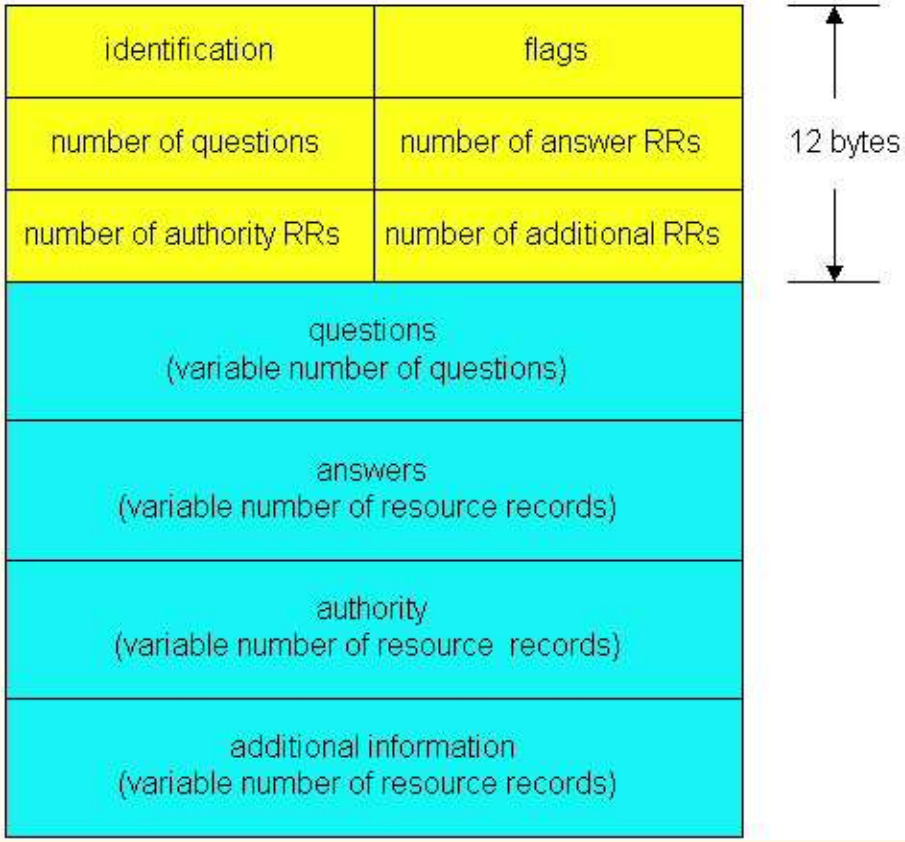
- Type=A
  - **name** is hostname
  - **value** is IP address
- Type=NS
  - name is domain (e.g. foo.com)
  - value is hostname of authoritative name server for this domain
- Type=CNAME
  - **name** is alias name for some "canonical" (the real) name `www.ibm.com` is really `servereast.backup2.ibm.com`
  - **value** is canonical name
- Type=MX
  - **value** is name of mailserver associated with **name**

DNS Protocol and Messages

DNS protocol: *query* and *reply* messages, both with the same *message format*.

msg header

- identification: 16 bit # for query, reply to query uses same #
- flags:
  - query or reply
  - recursion desired
  - recursion available
  - reply is authoritative



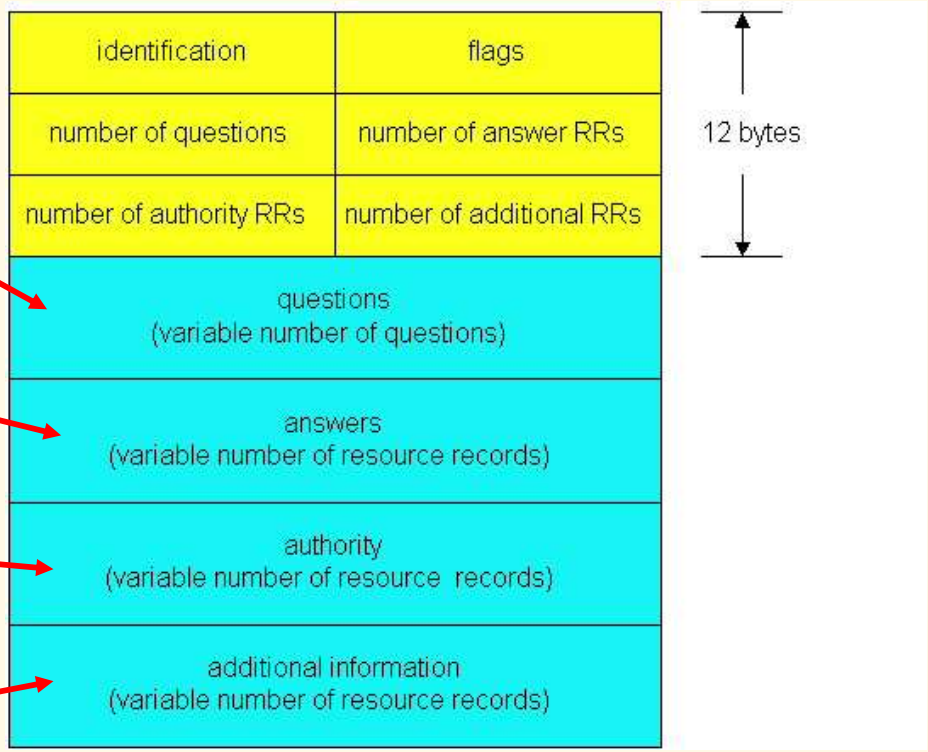
DNS Protocol and Messages

Name, type fields for a query

Resource records in response to query

Records for authoritative servers

Additional "helpful" info that may be used



Inserting records into DNS

- Example: new startup "Network Utopia"
  - How do people get IP address of your Web site?
  - How do they send you email?
- Register name `networkuptopia.com` at DNS *registrar* (e.g., Network Solutions)
  - provide names, IP addresses of authoritative name server (primary and secondary).
  - registrar inserts two RRs into `.com` TLD server:

```
(networkutopia.com, dns1.networkutopia.com, NS)
(dns1.networkutopia.com, 212.212.212.1, A)
```

- Create authoritative server Type A record for `www.networkuptopia.com`; Type MX record for `networkutopia.com` for mail.

DNS Summary

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